

Channel Estimation for Extremely Large-Scale MIMO: Far-Field or Near-Field?

Course: Principles of Wireless Communications

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1 Motivation

Extremely large-scale multiple-input multiple-output (XL-MIMO) uses hundreds of antennas to provide the array gain and spatial resolution needed for high-rate 6G links. Accurate channel state information is essential for precoding, yet the number of radio-frequency chains is much smaller than the number of antennas. Directly estimating every antenna coefficient would therefore require excessive pilot overhead.

Compressed sensing reduces this overhead by assuming that only a few paths are active in an angular Fourier dictionary. Cui and Dai show that this assumption breaks down when the XL-MIMO aperture places users or scatterers in the near field.

2 Far-Field and Near-Field Propagation

The boundary is approximated by the Rayleigh distance

$$Z = \frac{2D^2}{\lambda}, \quad (1)$$

where D is the array aperture and λ is the wavelength. Since Z grows quadratically with D , an XL-MIMO array can have a near-field region extending over a large part of a cell.

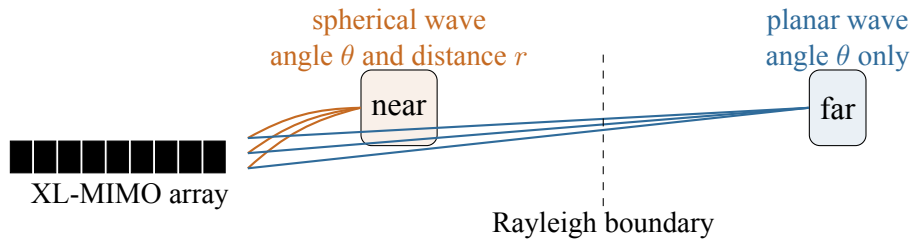


Figure 1: Far-field (planar-wave) and near-field (spherical-wave) propagation separated by the Rayleigh boundary.

Far-field phase changes linearly across the array, so one Fourier vector $\mathbf{a}(\theta)$ is sufficient. Near-field phase also depends on distance and requires $\mathbf{b}(\theta, r)$. An angle-only dictionary therefore spreads one physical near-field path over many bins.

3 Near-Field Channel Model

For a uniform linear array with antenna index $n = 0, 1, \dots, N-1$, spacing d , and wavenumber $k_c = 2\pi/\lambda$, the near-field steering vector is

$$\mathbf{b}(\theta, r) = \frac{1}{\sqrt{N}} \left[e^{-jk_c(r^{(0)}-r)}, \dots, e^{-jk_c(r^{(N-1)}-r)} \right]^T, \quad (2)$$

where

$$r^{(n)} = \sqrt{r^2 + \delta_n^2 d^2 - 2r\delta_n d \theta}, \quad \delta_n = \frac{2n - N + 1}{2}, \quad (3)$$

is the distance between the source and the n -th antenna located at $\delta_n d$, and $\theta \in [-1, 1]$ is the spatial angle. Unlike a Fourier vector, Equation (2) has a nonlinear phase progression that changes with both θ and r .

Figure 2 illustrates the system architecture. The N -element antenna array receives signals from scatterers in the near field. An analog combining matrix $\mathbf{A} \in \mathbb{C}^{N_{\text{RF}} \times N}$ reduces the N antenna outputs to N_{RF} RF chains, producing the observation $\bar{\mathbf{Y}}$. The polar-domain transform matrix $\mathbf{W}$ maps the sparse representation $\mathbf{H}^P$ back to the antenna domain. Because the number of RF chains is much smaller than N , compressed sensing is needed to recover the channel from few pilots.

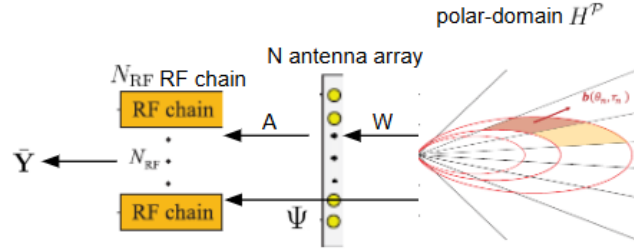


Figure 2: System model: the N -element array, analog combiner $\mathbf{A}$, RF chains, and polar-domain transform $\mathbf{W}$.

With a limited number of paths, the channel remains compressible even though it is not sparse in angle. The proposed representation is

$$\mathbf{h}_m = \mathbf{W} \mathbf{h}_m^P, \quad (4)$$

where every column of $\mathbf{W}$ is a sampled steering vector $\mathbf{b}(\theta_p, r_p)$ and $\mathbf{h}_m^P$ is sparse in the polar domain. The subscript m denotes a subcarrier; the path locations are shared across subcarriers, which enables simultaneous recovery.

4 From Energy Spread to Polar Sparsity

Figure 3 captures the key motivation. In the angular domain, a far-field path forms a narrow peak, while a near-field path occupies many bins. In the polar domain, each path is represented by an angle–distance pair and the channel again contains only a few significant coefficients. This restored sparsity is what makes low-pilot compressed sensing practical.

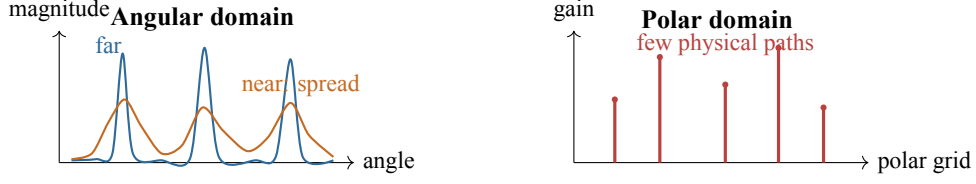


Figure 3: Energy spread and polar sparsity.

5 Design of the Polar-Domain Matrix W

The exact spherical distance makes direct grid design difficult. The Fresnel approximation separates its phase into

$$r^{(n)} - r \approx -\delta_n d\theta + \frac{\delta_n^2 d^2 (1 - \theta^2)}{2r}. \quad (5)$$

The linear term describes angle; the quadratic term contains the coupled quantity $(1 - \theta^2)/r$. This yields the two sampling rules below.

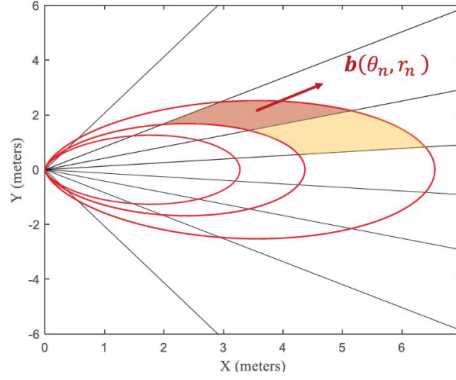


Figure 4: The polar-domain grid produced by the design: distance rings with uniformly sampled angles and steering vectors $\mathbf{b}(\theta_n, r_n)$.

Figure 4 shows the resulting polar-domain grid in physical coordinates. Each elliptical contour corresponds to a distance ring on which angles are sampled uniformly. The steering vector $\mathbf{b}(\theta_n, r_n)$ points to a specific angle–distance pair. Closer rings are spaced more densely because the sampling is uniform in inverse distance $1/r$, which ensures low mutual coherence between adjacent columns of W .

5.1 Angular Sampling on Distance Rings

Locations satisfying

$$\frac{1 - \theta^2}{r} = \frac{1}{\rho} \quad (6)$$

have the same quadratic phase and lie on a *distance ring* ρ . Within one ring, coherence is governed by the same linear phase as in a far-field Fourier dictionary. Therefore the angles are sampled uniformly:

$$\theta_n = \frac{2n - N + 1}{N}, \quad n = 0, 1, \dots, N - 1. \quad (7)$$

5.2 Nonuniform Distance Sampling

For vectors at the same angle, coherence depends on the difference between inverse distances. To keep adjacent columns below a chosen coherence threshold, the sampled distance at angle θ follows

$$r_s(\theta) = \frac{1}{s} Z_\Delta (1 - \theta^2), \quad s = 0, 1, 2, \dots, \quad (8)$$

where Z_Δ is determined by the array, wavelength, and desired coherence threshold. The case $s = 0$ denotes an infinite-distance far-field ring. Thus distance is sampled uniformly in inverse distance, not uniformly in metres.

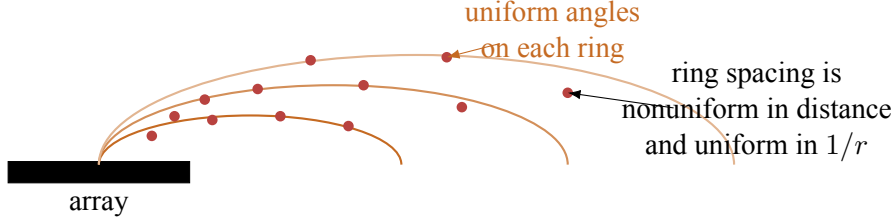


Figure 5: Conceptual distance-ring sampling used to construct $\mathbf{W}$.

5.3 Matrix Construction

Each ring produces

$$\mathbf{W}_s = [\mathbf{b}(\theta_0, r_{s,0}), \dots, \mathbf{b}(\theta_{N-1}, r_{s,N-1})], \quad (9)$$

and $\mathbf{W} = [\mathbf{W}_0, \dots, \mathbf{W}_{S-1}] \in \mathbb{C}^{N \times NS}$. The $s = 0$ far-field ring makes the Fourier dictionary a special case.

6 Near-Field Channel Estimation Algorithms

After analog combining, the noise is colored. The paper first applies a Cholesky-based pre-whitening transform and obtains the multi-subcarrier model

$$\overline{\mathbf{Y}} = \overline{\mathbf{\Phi}} \mathbf{H}^P + \overline{\mathbf{N}}, \quad \overline{\mathbf{\Phi}} = \mathbf{D}^{-1} \mathbf{A} \mathbf{W}. \quad (10)$$

The rows of $\mathbf{H}^P$ share a common sparse support because path angles and distances are common across subcarriers.

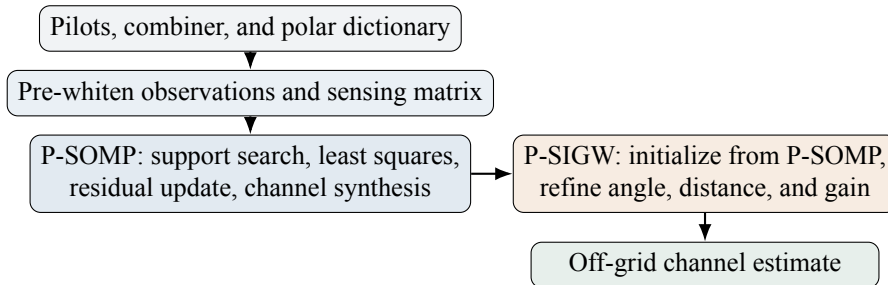


Figure 6: Combined workflow of the P-SOMP and P-SIGW algorithms.

6.1 On-Grid P-SOMP

P-SOMP extends simultaneous OMP from the angular dictionary to $\mathbf{W}$. It proceeds as follows:

1. construct $\mathbf{W}$ and pre-whiten the measurements;
2. initialize the residual and an empty support set;
3. select the polar atom with the largest summed correlation across subcarriers;
4. estimate gains by least squares and update the residual;
5. repeat for the detected paths and synthesize the channel.

Because $s = 0$ represents infinite distance, P-SOMP also works for far-field paths. Its weakness is grid mismatch: real angles and distances are continuous and rarely coincide exactly with sampled atoms.

6.2 Off-Grid P-SIGW

P-SIGW performs off-grid near-field channel estimation. Its optimization target is

$$\min_{\hat{\boldsymbol{\theta}}, \hat{\mathbf{r}}} \mathcal{L}(\hat{\boldsymbol{\theta}}, \hat{\mathbf{r}}) = -\text{Tr}\left\{\overline{\mathbf{Y}}^H \mathbf{P}(\hat{\boldsymbol{\theta}}, \hat{\mathbf{r}}) \overline{\mathbf{Y}}\right\}, \quad (11)$$

and the algorithm proceeds in three phases.

Phase 1: Initialization. Obtain the initial distances $\hat{\mathbf{r}}^0 = [\hat{r}_1^0, \hat{r}_2^0, \dots, \hat{r}_L^0]$ and angles $\hat{\boldsymbol{\theta}}^0 = [\hat{\theta}_1^0, \hat{\theta}_2^0, \dots, \hat{\theta}_L^0]$ from P-SOMP, then compute the polar-domain gain $\hat{\mathbf{G}} = \hat{\mathbf{H}}_r^{\mathcal{P}}$ and grid $\hat{\mathbf{r}}^0, \hat{\boldsymbol{\theta}}^0$.

Phase 2: Refinement. For $n = 1, 2, \dots, N_{\text{iter}}$, update angle and distance separately using Armijo backtracking line search:

$$\hat{\theta}^n = \hat{\theta}^{n-1} - l_1 \nabla_{\hat{\theta}} \mathcal{L}(\hat{\boldsymbol{\theta}}, \hat{\mathbf{r}}^{n-1})|_{\hat{\theta}=\hat{\theta}^{n-1}}, \quad (12)$$

$$\frac{1}{\hat{r}^n} = \frac{1}{\hat{r}^{n-1}} - l_2 \nabla_{\frac{1}{\hat{r}}} \mathcal{L}(\hat{\boldsymbol{\theta}}^n, \hat{\mathbf{r}}) = \hat{r}^{n-1}, \quad (13)$$

where l_1 and l_2 are the step lengths chosen by the Armijo rule. After each iteration, update the path gains via

$$\hat{\mathbf{G}}^{\text{opt}} = \tilde{\boldsymbol{\Psi}}^\dagger(\hat{\boldsymbol{\theta}}, \hat{\mathbf{r}}) \overline{\mathbf{Y}}. \quad (14)$$

Phase 3: Channel synthesis. Construct the estimated channel as $\hat{\mathbf{H}} = [\mathbf{b}(\hat{\theta}_1^n, \hat{r}_1^n), \mathbf{b}(\hat{\theta}_2^n, \hat{r}_2^n), \dots, \mathbf{b}(\hat{\theta}_L^n, \hat{r}_L^n)]$ and return $\hat{\mathbf{H}}$.

7 Simulation Results

The distance experiment uses $N = 256$ antennas, a 100 GHz carrier, SNR = 10 dB, and pilot length $P = 32$. The corresponding Rayleigh distance is about 100 m. Figure 7 redraws the principal trends from Fig. 10 of the paper rather than reproducing its image.

8 Interpretation

Inside the Rayleigh distance, angular-domain methods suffer severe model mismatch because a spherical path is not sparse in angle. Their NMSE improves only as distance increases and the wavefront becomes nearly planar. P-SOMP is much less sensitive to distance because its

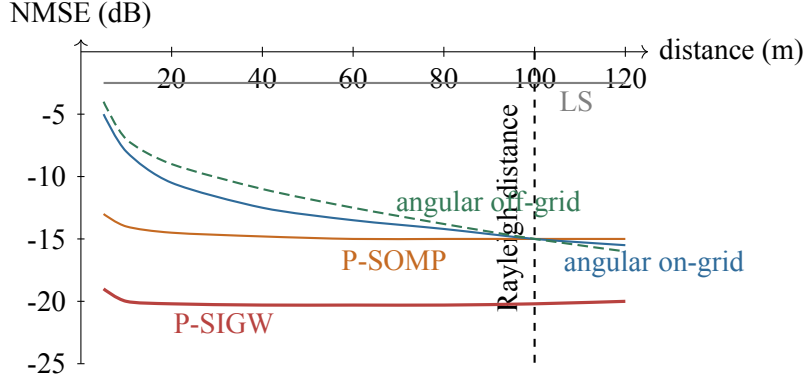


Figure 7: Trend-focused redraw of NMSE versus distance.

dictionary contains the required angle–distance atoms. P-SIGW is best because it also removes the polar-grid quantization error.

The other experiments in the paper support the same conclusion. As SNR increases, near-field polar methods continue improving while mismatched angular methods approach an error floor. In the far-field setting, the polar methods match the angular baselines and become slightly better at high SNR, because the Fresnel approximation is more accurate than the planar-wave one. Increasing pilot length improves all compressed-sensing estimators, but the gain is much larger when the representation matches the propagation physics.

Q&A

Question. While these two algorithms enhance performance, do they also introduce higher computational complexity compared to existing works? Furthermore, where does the primary complexity bound lie?

Answer.

1. *Do they introduce higher computational complexity?*

- **P-SOMP:** Overall complexity $\mathcal{O}(\hat{L}PN_{\text{RF}}NSM)$. Compared to the far-field on-grid algorithm SWOMP, its complexity is S times that of SWOMP, where S is the number of sampled distance rings.
- **P-SIGW:** Overall complexity $\mathcal{O}(\hat{L}PN_{\text{RF}}NSM + N_{\text{iter}}(P^2N_{\text{RF}}^2M + PN_{\text{RF}}M^2))$. Compared to the far-field off-grid algorithm SS-SIGW-OLS, **the refinement-stage complexity is exactly the same**. The extra cost comes solely from the initialization stage, which runs P-SOMP and thus carries the same S -fold overhead.

Summary: Although higher complexity is introduced, the non-uniform distance-sampling method is highly efficient— S is generally very small (4 to 6 in the simulations)—so this overhead is acceptable.

2. *Where does the primary complexity bound lie?*

- **P-SOMP bottleneck.** The dominant operation is computing the correlation matrix $\mathbf{\Gamma} = \bar{\mathbf{\Psi}}^H \mathbf{R}$ at each of the $\hat{L}$ iterations. Because XL-MIMO makes N very large and the polar-domain grid covers both angle and distance, the transform matrix $\bar{\mathbf{\Psi}} \in \mathbb{C}^{PN_{\text{RF}} \times NS}$ has NS columns. The bottleneck is therefore $\mathcal{O}(\hat{L}PN_{\text{RF}}NSM)$, the product of paths $\hat{L}$, pilot length P , RF chains N_{RF} , polar-domain grid size NS , and subcarriers M .
- **P-SIGW bottleneck.** The refinement stage dominates because $\hat{L}$ is small. Each iteration computes the gradients of the objective with respect to the angles $\hat{\boldsymbol{\theta}}$ and distances $\hat{\mathbf{r}}$, whose cost is $\mathcal{O}(P^2N_{\text{RF}}^2M + PN_{\text{RF}}M^2)$ per iteration. After N_{iter} iterations the refinement cost is $\mathcal{O}(N_{\text{iter}}(P^2N_{\text{RF}}^2M + PN_{\text{RF}}M^2))$. The two terms grow quadratically with the effective observation dimension PN_{RF} and with the subcarrier count M , respectively.